

TECHNICAL SKILLS

Control Systems and Robotics: ROS/ROS2, State-Space Control, LQR, PID, Optimal Control, MPC, SLAM, RL
Programming: Python, C++, PyTorch, SQL, MATLAB, Git
Engineering Tools: Isaacsim, Mujoco, SIMULINK, SOLIDWORKS, ANSYS, MSC Adams

EDUCATION

MS Robotics and Autonomous Systems (Systems Engineering) <i>Arizona State University, Tempe, AZ</i>	08/2025 – 05/2027 GPA: 3.89
BS Mechanical Engineering <i>Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)</i>	08/2018 – 02/2023 Topi, Pakistan

PROFESSIONAL EXPERIENCE

Student Researcher <i>GHOST Lab, CHART, Arizona State University</i> - Calibrated a motion capture system to detect the real-time position of two rigid bodies. - Developed ROS nodes for real-time control of the robot and feedback collection. - Used Extended Kalman Filter (EKF) to reduce noise in motion capture data. - Designed a model-based, receding horizon controller for multi-agent navigation. - Incorporated Bayesian intent inference to enable belief-aware decision making.	08/2025 – Ongoing Tempe, AZ
tkxel (Software and AI Consultancy) <i>Data Scientist</i> - Architected and built a web backend for AI solutions targeting enterprise customers. - Trained a neural network in PyTorch to estimate software engineering effort with a 15% error. - Used Bayesian inference with LLMs to audit cybersecurity compliance records of corporations.	12/2023 – 07/2025 Lahore, Pakistan
Maqsood Labs <i>Associate Software Engineer</i> Built a MAVLink-based offboard control backend for PX4-compatible quadrotor UAVs.	02/2023 – 12/2023 Lahore, Pakistan

PROJECTS

Quadruped RL with 2-DoF Spine <i>Personal Project</i> - Modified a Unitree Go2 to make a 2 joint spine - Developed an RL policy to control the modified Go2 to navigate tight corners in a narrow-hallway. - Designed the action and observation spaces using IMU + joint states and relative obstacle/goal pose. - Trained the policy in Isaac Sim and iterating environment dynamics and constraints.	02/2026 – Present Tempe, AZ
LQR Control for Motor Position Control <i>Arizona State University</i> - Built an LQR position controller in SIMULINK and ported to Arduino. - Implemented state-feedback control with IIR velocity filtering to reject encoder noise. - Developed and analyzed an LQR cart-pole controller in ROS2 + Gazebo under stochastic earthquake disturbances, tuning Q/R matrices to balance stability and control effort. - Evaluated system performance using pole angle deviation, cart displacement, and control effort metrics.	11/2025 – 11/2025 Tempe, AZ
Design and Control of Throat Swab Sampling Robot <i>Final Year Project (Capstone Project)</i> - Placed 3rd out of 25 teams in the industrial open house. - Designed a 6 DOF parallel manipulator in SOLIDWORKS and ran dynamic analysis in MSC Adams. - Simulated inverse kinematics in MATLAB to compute joint angles. - Used Jacobian matrix to calculate required joint torques. - Developed ROS nodes on a Raspberry Pi and tuned control parameters on the real robot.	07/2021 – 05/2022 Topi, Pakistan